The Hellings-Downs curve for a Gaussian source

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This paper is devoted to the derivation of the Hellings-Downs curve for the case of a Gaussian source of the Gravitational Wave Background.

1 Problem Statement

Consider the spherical coordinate system (ϕ, θ, r), where ϕ is longitude, θ is colatitude and r is radius. Then the orthonormal basis in the cartesian coordinate system will look like:

$$\hat{\Omega} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)\,, \tag{1a}$$
$$\hat{n} = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta), \tag{1b}$$
$$\hat{m} = (\sin\phi, -\cos\phi, 0), \tag{1c}$$

where $(\hat{\Omega}, \hat{n}, \hat{m}) \equiv (\hat{r}, \hat{\theta}, -\hat{\phi})$. Since $\bar{r} = r \cdot \hat{r}$, and the point on the unit sphere has the value $r = 1$, we can describe the unit sphere with a single vector $\hat{\Omega} = \hat{r}$.

We also know that gravitational waves have two polarization, which are called "plus" and "cross", and are described by the corresponding tensors:

$$e^{+}_{\alpha\beta}(\hat{\Omega}) = \hat{m}_\alpha \hat{m}_\beta - \hat{n}_\alpha \hat{n}_\beta, \tag{2a}$$
$$e^{\times}_{\alpha\beta}(\hat{\Omega}) = \hat{m}_\alpha \hat{n}_\beta + \hat{n}_\alpha \hat{m}_\beta \tag{2b}$$

Then, frequency shift of pulsar radiation $z = \Delta\nu/\nu_0$ in Fourier space:

$$\tilde{z}(f, \hat{\Omega}) = \left(e^{-i2\pi f L(1+\hat{\Omega}\cdot\hat{p})} - 1\right) h(f, \hat{\Omega}) F_{\hat{p}}(f, \hat{\Omega}), \tag{3}$$

where $\hat{p}$ is the unit vector in the direction of the pulsar, L is the distance to the pulsar, $h(f, \hat{\Omega})$ is spectrum of gravitational waves and $F_{\hat{p}}(\hat{\Omega})$ is pulsar response:

$$F_{\hat{p}}(\hat{\Omega}) = e_{\alpha\beta}(\hat{\Omega}) \frac{1}{2} \frac{\hat{p}^\alpha \hat{p}^\beta}{1 + \hat{\Omega}\cdot\hat{p}}. \tag{4}$$

Here we use a complex representation of the polarization tensor and spectrum, and in the case where the GWB is unpolarized, we have:

$$h(f, \hat{\Omega}) = h_+(f, \hat{\Omega}) + i \cdot h_\times(f, \hat{\Omega}) \tag{5}$$
$$e_{\alpha\beta}(\hat{\Omega}) = e^{+}_{\alpha\beta}(\hat{\Omega}) + i \cdot e^{\times}_{\alpha\beta}(\hat{\Omega}) \tag{6}$$

Since the Hellings-Downs curve relates correlations between signals from two pulsars, we defined the Hellings-Downs curve as:

$$\Gamma[\mathcal{P}](\hat{p}, \hat{q}) = \frac{3}{4\pi} \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) \cdot \mathcal{P}(\hat{\Omega}), \tag{7}$$

where $\mathcal{P}(\hat{\Omega})$ is the power spectrum on the unit sphere, and with the frequency power spectrum $\mathcal{H}(f)$ defined as:

$$|h(f, \hat{\Omega})|^2 = \mathcal{H}(f) \cdot \mathcal{P}(\hat{\Omega}), \tag{8}$$

and $\mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q})$ is the integral kernel, which is defined as:

$$\mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) = \mathcal{R}\left[F^*_{\hat{p}}(\hat{\Omega}) \cdot F_{\hat{q}}(\hat{\Omega})\right]. \tag{9}$$

And in our case, we assume that the power spectrum is Gaussian and looks like this:

$$\mathcal{P}(\hat{\Omega}) = \exp\left[-\frac{(\hat{\Omega} - \hat{\Omega}_0)^2}{2\sigma^2}\right], \tag{10}$$

where $\hat{\Omega}_0$ is the center of the source and σ is the width of the Gaussian. Converting the numerator:

$$(\hat{\Omega} - \hat{\Omega}_0)^2 = \hat{\Omega}^2 + \hat{\Omega}_0^2 - 2\hat{\Omega}\cdot\hat{\Omega}_0 = 1 + 1 - 2\hat{\Omega}\cdot\hat{\Omega}_0$$

$$(\hat{\Omega} - \hat{\Omega}_0)^2 = 2(1 - \hat{\Omega}\cdot\hat{\Omega}_0),$$

we have:

$$\mathcal{P}(\hat{\Omega}) = \exp\left[-\frac{2(1 - \hat{\Omega}\cdot\hat{\Omega}_0)}{2\sigma^2}\right],$$

$$\mathcal{P}(\hat{\Omega}) = \exp\left[-\frac{1}{\sigma^2}\right] \cdot \exp\left[\frac{\hat{\Omega}\cdot\hat{\Omega}_0}{\sigma^2}\right].$$

Let $\kappa = 1/\sigma^2$, and as a result we have:

$$\mathcal{P}(\hat{\Omega}) = \exp\left(-\kappa\right) \cdot \exp\left(\kappa\hat{\Omega}\cdot\hat{\Omega}_0\right). \tag{11}$$

And to find the Hellings-Downs curve for a Gaussian source, we need to calculate:

$$\Gamma^{(g)}(\hat{p}, \hat{q}) = \frac{3e^{-\kappa}}{4\pi} \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) \cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0} \tag{12}$$

2 Integral in Coordinates

Firstly, let's represent the integral kernel $\mathcal{K}$ in vector form:

$$\mathcal{K} = \mathcal{R}\left[e^*_{\alpha\beta}(\hat{\Omega})\frac{1}{2}\frac{\hat{p}^\alpha\hat{p}^\beta}{1+\hat{\Omega}\cdot\hat{p}}\cdot e_{\mu\nu}(\hat{\Omega})\frac{1}{2}\frac{\hat{q}^\mu\hat{q}^\nu}{1+\hat{\Omega}\cdot\hat{q}}\right],$$

$$\mathcal{K} = \frac{1}{4}\frac{\hat{p}^\alpha\hat{p}^\beta\hat{q}^\mu\hat{q}^\nu}{(1+\hat{\Omega}\cdot\hat{p})(1+\hat{\Omega}\cdot\hat{q})}\mathcal{R}\left[e^*_{\alpha\beta}(\hat{\Omega})\cdot e_{\mu\nu}(\hat{\Omega})\right],$$

where:

$$\mathcal{R}\left[e^*_{\alpha\beta}(\hat{\Omega})\cdot e_{\mu\nu}(\hat{\Omega})\right] = e^+_{\alpha\beta}(\hat{\Omega})e^+_{\mu\nu}(\hat{\Omega}) + e^\times_{\alpha\beta}(\hat{\Omega})e^\times_{\mu\nu}(\hat{\Omega}),$$

then:

$$\mathcal{K} = \frac{1}{4}\frac{\hat{p}^\alpha\hat{p}^\beta\hat{q}^\mu\hat{q}^\nu\left[e^+_{\alpha\beta}(\hat{\Omega})e^+_{\mu\nu}(\hat{\Omega}) + e^\times_{\alpha\beta}(\hat{\Omega})e^\times_{\mu\nu}(\hat{\Omega})\right]}{(1+\hat{\Omega}\cdot\hat{p})(1+\hat{\Omega}\cdot\hat{q})}. \tag{13}$$

Consider individual tensor convolutions:

$$\hat{v}^\alpha\hat{v}^\beta e^+_{\alpha\beta}(\hat{\Omega}) = (\hat{v}\cdot\hat{m})^2 - (\hat{v}\cdot\hat{n})^2, \tag{14a}$$
$$\hat{v}^\alpha\hat{v}^\beta e^\times_{\alpha\beta}(\hat{\Omega}) = 2(\hat{v}\cdot\hat{m})(\hat{v}\cdot\hat{n}). \tag{14b}$$

Let's denote $P_a = (\hat{p}\cdot\hat{a})$ and $Q_a = (\hat{q}\cdot\hat{a})$, then:

$$\mathcal{K} = \frac{1}{4}\frac{(P_m^2 - P_n^2)\cdot(Q_m^2 - Q_n^2) + 4P_mP_nQ_mQ_n}{(1+P_\Omega)(1+Q_\Omega)}. \tag{15}$$

Note that since $(\hat{m},\hat{n},\hat{\Omega})$ is an orthonormal basis, and $|\hat{p}| = 1$, $|\hat{q}| = 1$, it means that:

$$P_n^2 = 1 - P_m^2 - P_\Omega^2,$$
$$Q_n^2 = 1 - Q_m^2 - Q_\Omega^2.$$

Then:

$$\frac{(P_n^2 - P_m^2)}{1+P_\Omega} = \frac{1 - 2P_m^2 - P_\Omega^2}{1+P_\Omega} = (1-P_\Omega) - \frac{2P_m^2}{1+P_\Omega},$$
$$\frac{(Q_n^2 - Q_m^2)}{1+Q_\Omega} = \frac{1 - 2Q_m^2 - Q_\Omega^2}{1+Q_\Omega} = (1-Q_\Omega) - \frac{2Q_m^2}{1+Q_\Omega}.$$

As a result, we can divide the integral kernel into four parts:

$$\mathcal{K}(\hat{\Omega},\hat{p},\hat{q}) = \frac{1}{4}(\mathcal{K}_I + \mathcal{K}_J + \mathcal{K}_K + \mathcal{K}_H), \tag{16}$$

where:

$$\mathcal{K}_I = (1-P_\Omega)(1-Q_\Omega), \tag{17a}$$
$$\mathcal{K}_J = -2\cdot\frac{P_m^2(1-Q_\Omega)}{(1+P_\Omega)}, \tag{17b}$$
$$\mathcal{K}_K = -2\cdot\frac{Q_m^2(1-P_\Omega)}{(1+Q_\Omega)}, \tag{17c}$$
$$\mathcal{K}_H = 4\cdot\frac{P_mQ_m(P_mQ_m + P_nQ_n)}{(1+P_\Omega)(1+Q_\Omega)}, \tag{17d}$$

and we can divide the integral into the same four parts:

$$\Gamma^{(g)}(\hat{p},\hat{q}) = \frac{3e^{-\kappa}}{4\pi}\cdot\frac{1}{4}(I + J + K + H), \tag{18}$$

where:

$$I = \int_{S^2} d\hat{\Omega}\cdot\mathcal{K}_I\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0} \tag{19a}$$
$$J = \int_{S^2} d\hat{\Omega}\cdot\mathcal{K}_J\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0} \tag{19b}$$
$$K = \int_{S^2} d\hat{\Omega}\cdot\mathcal{K}_K\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0} \tag{19c}$$
$$H = \int_{S^2} d\hat{\Omega}\cdot\mathcal{K}_H\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0} \tag{19d}$$

Let's define a convenient coordinate system. As we can see, all integrals include a term containing $(\hat{\Omega}\cdot\hat{\Omega}_0)$ in the exponent, and to simplify this term, we take $\hat{\Omega}_0 \equiv (0,0,1)$, because then $(\hat{\Omega}\cdot\hat{\Omega}_0) = \cos\theta$. For the pulsar vectors, we have:

$$\hat{p} = (\sin\alpha, 0, \cos\alpha) \tag{20a}$$
$$\hat{q} = (\sin\beta\cos\gamma, \sin\beta\sin\gamma, \cos\beta) \tag{20b}$$

where α, β are the angles between the source center and the pulsars, and γ is the angle between the directions to the pulsars from the source center. Now we can calculate all the scalar products:

$$P_\Omega = \sin\alpha\sin\theta\cos\phi + \cos\alpha\cos\theta,$$
$$Q_\Omega = \sin\beta\sin\theta\cos(\phi-\gamma) + \cos\beta\cos\theta,$$
$$P_n = \sin\alpha\cos\theta\cos\phi - \cos\alpha\sin\theta,$$
$$Q_n = \sin\beta\cos\theta\cos(\phi-\gamma) - \cos\beta\sin\theta$$
$$P_m = \sin\alpha\sin\phi,$$
$$Q_m = \sin\beta\sin(\phi-\gamma);$$

and define scalar product between $\hat{p}$ and $\hat{q}$ as:

$$\cos\xi = \sin\alpha\sin\beta\cos\gamma + \cos\alpha\cos\beta, \tag{21}$$

where ξ is the angle between $\hat{p}$ and $\hat{q}$.

Also, let's highlight the dependence of the coefficients on ϕ:

$$P_\Omega = a\cos\phi + b, Q_\Omega = c\cos(\phi-\gamma) + d, \tag{22a}$$
$$P_n = h\cos\phi - s, Q_n = k\cos(\phi-\gamma) - t, \tag{22b}$$
$$P_m = f\sin\phi, Q_m = g\sin(\phi-\gamma); \tag{22c}$$

where:

$$a = \sin\alpha\sin\theta, b = \cos\alpha\cos\theta, \tag{23a}$$
$$c = \sin\beta\sin\theta, d = \cos\beta\cos\theta, \tag{23b}$$
$$h = \sin\alpha\cos\theta, s = \cos\alpha\sin\theta, \tag{23c}$$
$$k = \sin\alpha\cos\theta, t = \cos\alpha\sin\theta, \tag{23d}$$
$$f = \sin\alpha, g = \sin\beta, \tag{23e}$$

3 Calculation of Integrals

All integrals are calculated first in longitude and then in latitude, and in all of them, after the first integration, we get an integral only in $\cos\theta$, and the integral can be replaced by an integral from -1 to $+1$ in x. Let's calculate the integral of I using these steps:

$$I = \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}_I \cdot e^{\kappa \hat{\Omega}\cdot\hat{\Omega}_0}$$

$$I = \int_0^\pi \sin\theta \cdot d\theta \cdot e^{\kappa\cdot\cos\theta} \int_0^{2\pi} d\phi \cdot \mathcal{K}_I \tag{24}$$

$$I_\phi = \int_0^{2\pi} d\phi \cdot \mathcal{K}_I \tag{25}$$

The result for I_ϕ is:

$$I_\phi = 2\pi \cdot \left[(b-1)(d-1) + \frac{1}{2} \cdot ac\cos\gamma\right] \tag{26}$$

and after highlighting the dependency on $\cos\theta$:

$$\begin{aligned} I_\phi = 2\pi \cdot \Big[&\cos^2\theta \cdot \left(\frac{3}{2}\cos\alpha\cos\beta - \frac{1}{2}\cos\xi\right) \\ &- \cos\theta \cdot (\cos\alpha + \cos\beta) \\ &+ \left(1 + \frac{1}{2}\cos\xi - \frac{1}{2}\cos\alpha\cos\beta\right)\Big] \end{aligned} \tag{27}$$

Then we can rewrite the final integral as an integral in x:

$$\begin{aligned} I &= \int_0^\pi (-1) \cdot d(\cos\theta) \cdot e^{\kappa\cdot\cos\theta} \cdot I_\phi(\cos\theta) \\ &= \int_{-1}^1 dx \cdot I_\phi(x) \cdot e^{\kappa x}. \end{aligned} \tag{28}$$

As a result, the integral I will be equal to:

$$\begin{aligned} I = 4\pi\Big[&\frac{\sinh\kappa}{\kappa} + \frac{3(\kappa\cosh\kappa - \sinh\kappa)}{\kappa^3} \cdot \frac{\cos\xi}{3} \\ &+ \left(\frac{\sinh\kappa}{\kappa} - \frac{3(\kappa\cosh\kappa - \sinh\kappa)}{\kappa^3}\right) \cdot \cos\alpha\cos\beta \\ &- \frac{\kappa\cosh\kappa - \sinh\kappa}{\kappa^2} \cdot (\cos\alpha + \cos\beta)\Big] \end{aligned} \tag{29}$$

Let's calculate the integral J using the same algorithm:

$$J = \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}_J \cdot e^{\kappa \hat{\Omega}\cdot\hat{\Omega}_0}$$

$$J = \int_0^\pi \sin\theta \cdot d\theta \cdot e^{\kappa\cdot\cos\theta} \int_0^{2\pi} d\phi \cdot \mathcal{K}_J \tag{30}$$

$$J_\phi = \int_0^{2\pi} d\phi \cdot \mathcal{K}_J \tag{31}$$

The result for J_ϕ is:

$$\begin{aligned} J_\phi = 4\pi f^2 \cdot \Big[&\frac{(b+1) - \sqrt{(b+1)^2 - a^2}}{a^2} \cdot (d-1) \\ &+ \left(\frac{(b+1) - \sqrt{(b+1)^2 - a^2}}{a^2}\right)^2 \cdot \frac{ac\cos\gamma}{2}\Big] \end{aligned} \tag{32}$$

and after highlighting the dependency on $\cos\theta$:

$$\begin{aligned} J_\phi = 4\pi \cdot \Big[&\frac{1 + \cos\alpha\cos\theta - |\cos\alpha + \cos\theta|}{1 - \cos^2\theta} \cdot (\cos\beta\cos\theta - 1) \\ &+ \frac{(1 + \cos\alpha\cos\theta - |\cos\alpha + \cos\theta|)^2}{1 - \cos^2\theta} \cdot \frac{\cos\xi - \cos\alpha\cos\beta}{2(1 - \cos\alpha^2)}\Big] \end{aligned} \tag{33}$$

Then we can rewrite the final integral as an integral in x:

$$\begin{aligned} J &= \int_0^\pi (-1) \cdot d(\cos\theta) \cdot e^{\kappa\cdot\cos\theta} \cdot J_\phi(\cos\theta) \\ &= \int_{-1}^1 dx \cdot J_\phi(x) \cdot e^{\kappa x} \end{aligned} \tag{34}$$

As a result, the integral J will be equal to:

$$\begin{aligned} J = 4\pi\Big[&\frac{(1 - \cos\alpha) \cdot (1 + \cos\alpha + \cos\beta + \cos\xi)}{1 + \cos\alpha} \cdot \\ &e^{-\kappa} \cdot (\mathrm{Ei}[\kappa(1 - \cos\alpha)] - \mathrm{Ei}[2\kappa]) \\ &+ \frac{(1 + \cos\alpha) \cdot (1 - \cos\alpha - \cos\beta + \cos\xi)}{1 - \cos\alpha} \cdot \\ &e^{\kappa} \cdot (\mathrm{Ei}[-\kappa(1 + \cos\alpha)] - \mathrm{Ei}[-2\kappa]) \\ &+ (1 - \cos\alpha) \cdot \left(\cos\beta + \frac{1}{2}(\cos\alpha\cos\beta + \cos\xi)\right) \cdot \\ &\left(\frac{\sinh\kappa}{\kappa} + \cos\alpha \cdot \frac{\cosh\kappa - \exp(-\kappa\cdot\cos\alpha)}{\kappa\cdot\cos\alpha}\right) \cdot \frac{1}{1 + \cos\alpha} \\ &- (1 + \cos\alpha) \cdot \left(\cos\beta - \frac{1}{2}(\cos\alpha\cos\beta + \cos\xi)\right) \cdot \\ &\left(\frac{\sinh\kappa}{\kappa} - \cos\alpha \cdot \frac{\cosh\kappa - \exp(-\kappa\cdot\cos\alpha)}{\kappa\cdot\cos\alpha}\right) \cdot \frac{1}{1 - \cos\alpha}\Big] \end{aligned} \tag{35}$$

And since the integral K has the same structure, we can write the final equation for K by simply replacing

$\cos\alpha$ and $\cos\beta$ in the equation for the integral J:

$$
\begin{aligned}
K = 4\pi\Big[& \frac{(1-\cos\beta)\cdot(1+\cos\alpha+\cos\beta+\cos\xi)}{1+\cos\beta}\cdot \\
& e^{-\kappa}\cdot(\mathrm{Ei}[\kappa(1-\cos\beta)]-\mathrm{Ei}[2\kappa]) \\
& + \frac{(1+\cos\beta)\cdot(1-\cos\alpha-\cos\beta+\cos\xi)}{1-\cos\beta}\cdot \\
& e^{\kappa}\cdot(\mathrm{Ei}[-\kappa(1+\cos\beta)]-\mathrm{Ei}[-2\kappa]) \\
& + (1-\cos\beta)\cdot\left(\cos\alpha+\frac{1}{2}(\cos\alpha\cos\beta+\cos\xi)\right)\cdot \\
& \left(\frac{\sinh\kappa}{\kappa}+\cos\beta\cdot\frac{\cosh\kappa-\exp(-\kappa\cdot\cos\beta)}{\kappa\cdot\cos\beta}\right)\cdot\frac{1}{1+\cos\beta} \\
& - (1+\cos\beta)\cdot\left(\cos\alpha-\frac{1}{2}(\cos\alpha\cos\beta+\cos\xi)\right)\cdot \\
& \left(\frac{\sinh\kappa}{\kappa}-\cos\beta\cdot\frac{\cosh\kappa-\exp(-\kappa\cdot\cos\beta)}{\kappa\cdot\cos\beta}\right)\cdot\frac{1}{1-\cos\beta}\Big]
\end{aligned}
\tag{36}
$$